



AI · SEO

COURSE SLIDES · NON-WSQ

Generative AI for SEO

Course Code: C11

Tertiary Infotech Academy Pte Ltd · UEN
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COURSE ADMINISTRATION

Welcome and Outcomes

About the Trainer



Your Trainer

General Trainer template — to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer · Tertiary Infotech
Academy Pte Ltd

ROLE

Principal Trainer and course facilitator

SPECIALISMS

Generative AI, SEO, digital marketing and workflow design

DELIVERY

Professional development and workplace application

CONTACT

enquiry@tertiaryinfotech.com

Let's Know Each Other

1

Role

Your organisation, audience and current digital-marketing responsibility.

2

Search experience

One SEO task you do manually today.

3

AI experience

One generative-AI tool or prompt pattern you have tried.

4

Learning goal

One website outcome you want to improve responsibly.

Ground Rules

1

Participate

Challenge assumptions and contribute evidence.

2

Respect

Disagree with ideas, not people.

3

Confidentiality

Use public or course data only; do not disclose live incident information.

4

Devices

Silent mode; step out quietly if needed.

5

Time

Return from breaks promptly.

6

Attendance

Attend both days to get the most from the course.

Course Outline — 1 Day

Morning

- Morning — Strategy and discovery
 - Topic 1: SEO, generative AI and agentic workflows
 - Lab 1: baseline audit
 - Topic 2: keywords, intent, clusters and briefs
 - Lab 3: keyword universe

Afternoon

- Afternoon — Create, optimise and measure
 - Topic 3: blog, image, video and social assets
 - Topic 4: on-page SEO and Google AI features
 - Topic 5: Search Console, GA4 and governance
 - Labs 5, 7, 9; recap, Q&A and feedback

Learning Outcomes

1

LO1

Evaluate SEO and internet marketing strategies of a website.

2

LO2

Manage keyword research in alignment with SEO objectives.

3

LO3

Provide on-page SEO strategies and recommendations.

4

LO4

Manage off-page SEO strategies and performance across multiple channels.

5

LO5

Monitor Google Analytics and Search Console reports to evaluate SEO performance.

Skills Framework Alignment

Non-WSQ commercial short course

- Search Engine Optimisation (SEO)
- Proficiency Level 4

Knowledge K1–K10

- strategy · keywords · ranking · search methods · social · pages · performance

Abilities A1–A6

- research · recommend · manage · monitor · partner · evaluate

Access the Hands-On Labs

<https://github.com/tertiarycourses/C11-Generative-AI-for-SEO>

Clone

Clone the repository, then open the labs folder.

Download ZIP

Use GitHub Code → Download ZIP when Git is unavailable.

One shared site

Every activity applies its prompts and evidence to the same live EcoDesk website.

One EcoDesk Website · Ten Progressive Activities

Fictional learner website for SEO practice · Prices and business details are synthetic

ecodesk
work better · waste less

Solutions Products Insights FAQ [Get a workspace plan](#)

WORKSPACE PLANNING FOR SINGAPORE TEAMS

Turn every square metre into a better workday.

Ergonomic desks, adaptable team bundles and a practical plan—built around how your people actually work.

[Get my free workspace plan](#) [Compare solutions](#)

FREE 2026 GUIDE

The Smarter Workplace Buying Guide

Get the 12-point planning checklist, bundle comparison and budget worksheet used in this training demo.

Name

Work email

Team size

[Chat](#)

Use the same site for:

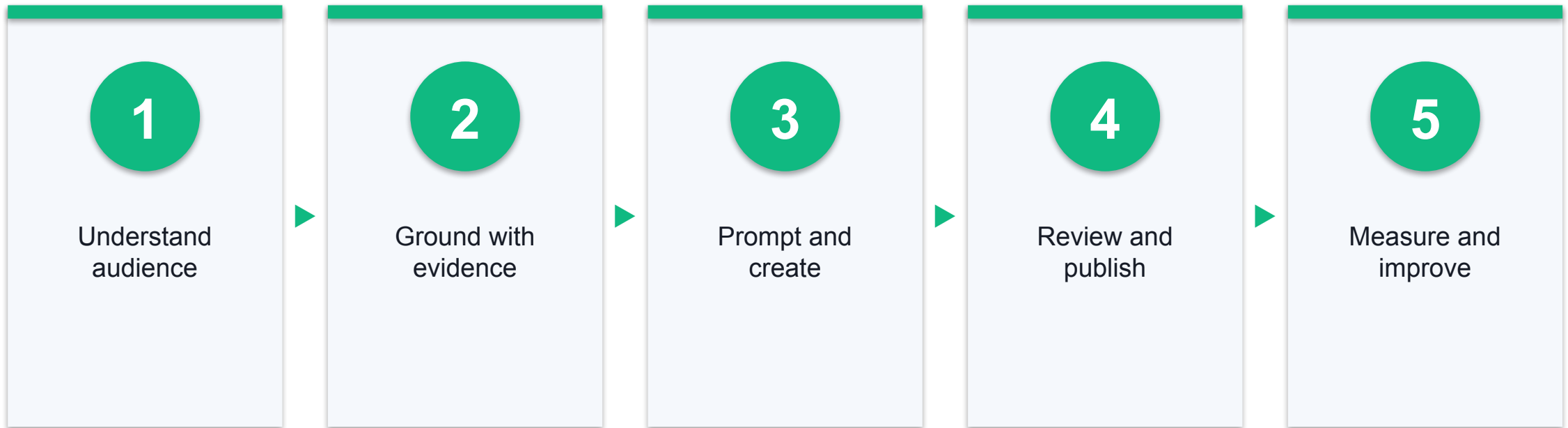
- audit and strategy
- keywords and content
- on-page and AI search
- measurement and agents

[OPEN LIVE ECODESK WEBSITE ↗](#)

CORE MODEL

**Intent · Evidence · Experience · Authority ·
Learning**

The AI SEO lifecycle



Evidence hierarchy

1

Primary guidance

Official search, analytics, standards or platform documentation.

2

First-party evidence

Search Console, GA4, customer language, site content and business outcomes.

3

Training assumption

A simulated input used only for learning and clearly labelled.

4

AI inference

A hypothesis that must be verified before action.

COURSE MODEL

Prompt quality cannot rescue missing evidence.

Ground the model, make uncertainty visible and keep a human accountable for publication.



LEARNING UNIT 01 · A6 · K1 · K6

Fundamentals of SEO and Generative AI

Search systems · strategy · responsible AI · agentic workflows

Key Concepts — Fundamentals of SEO and Generative AI

1

Search

Earn discoverability by helping people and search systems understand useful content.

2

Generative AI

Accelerate research and creation while humans own evidence, intent and approval.

3

Strategy

Connect audience needs, business outcomes, content assets, authority and measurement.

4

Governance

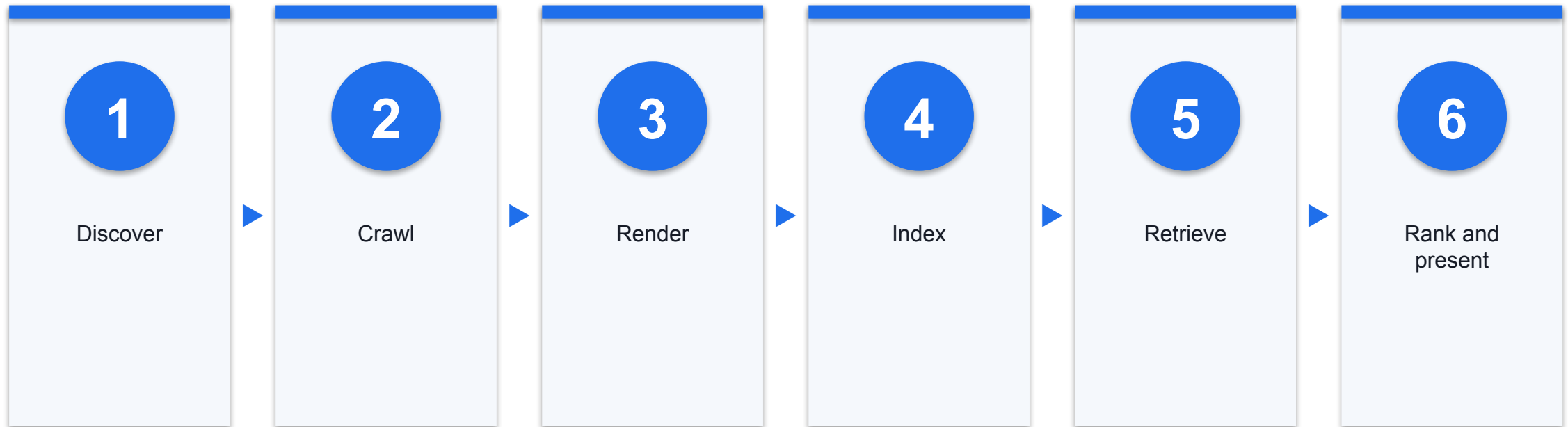
Set data, fact-check, brand, copyright, disclosure and publishing controls.

TOPIC 01 · A6 · K1 · K6

SEO is a system of useful signals, not a bag of tricks.

AI increases execution speed; strategy decides whether that speed creates value or spam.

How modern search finds and serves content



The SERP is now a portfolio

1

Organic links

Classic results with titles, snippets and rich features.

2

AI Overviews / AI Mode

Generated responses grounded in indexed sources and query fan-out.

3

Local, image and video

Intent-specific surfaces with their own eligibility signals.

4

Paid and shopping

Commercial inventory distinct from organic ranking.

Traditional SEO and generative-search visibility

Perspective A

- Indexable, useful pages
- Intent and relevance
- Authority and links
- Page experience and technical health

Perspective B

- Clear answer units
- Unique evidence and first-party experience
- Entity consistency
- Extractable text plus helpful images/video

White-hat and black-hat choices

1

People-first

Create original value for a defined audience.

2

Transparent

Use accurate metadata and visible content.

3

Manipulative

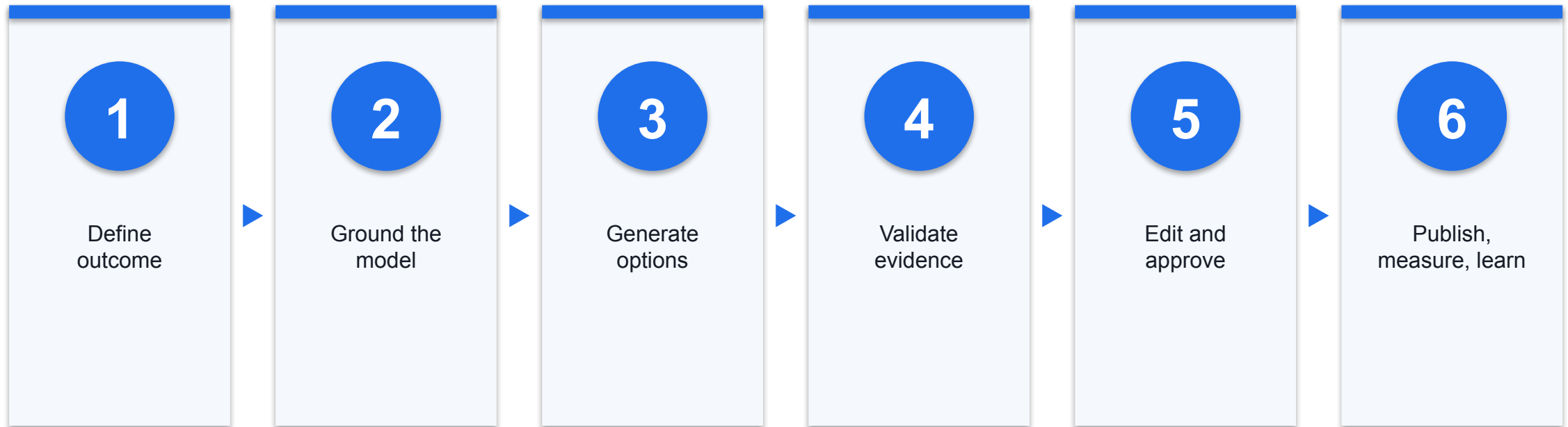
Cloaking, doorway pages, link schemes and hidden text.

4

Scaled abuse

Many unoriginal AI pages made mainly to manipulate rankings.

AI SEO operating cycle



AI SEO strategy: SIGNAL

1

S — Search intent

Specify audience, job and decision stage.

2

I — Information advantage

Add original examples, expertise and proof.

3

G — Generative workflow

Use reusable prompts, inputs and quality gates.

4

N — Network authority

Build internal links, earned mentions and credible distribution.

5

A — Accessibility

Ensure crawl, render, mobile and media accessibility.

6

L — Learning loop

Measure outcomes and refresh from evidence.

Three roles for AI

Copilot

- Human directs each task and approves the output.

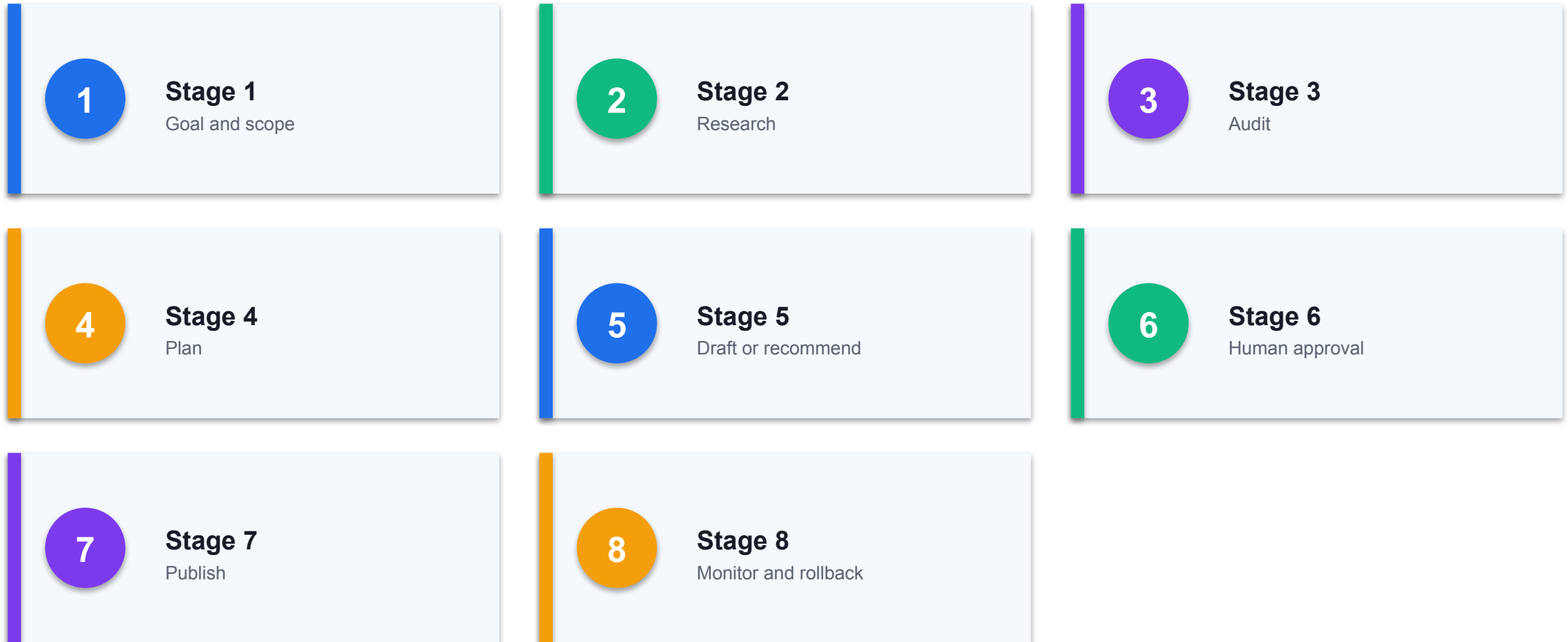
Workflow

- Automation passes structured inputs between tools.

Agent

- System plans and acts within a bounded goal, tools and approval policy.

Agentic SEO with control points



Human-in-the-loop risk tiers

1

Low

Summarise exports, cluster supplied keywords, flag missing metadata.

2

Medium

Draft briefs, internal links, schema and social variants for review.

3

High

Publish, change templates, edit robots/canonicals or contact external sites.

4

Control

Least privilege, preview, approvals, logs, limits and rollback.

Prompt anatomy: CONTEXT

1

C — Context

Business, audience, page, product and constraints.

2

O — Objective

One measurable task and target outcome.

3

N — Necessary inputs

Verified facts, keyword data, page copy and links.

4

T — Task steps

Analysis sequence, checks and decision rules.

5

E — Evidence

Citations, uncertainty labels and claim ledger.

6

X — eXclusions

Prohibited claims, tactics, private data and assumptions.

7

T — Template

Exact output fields, length limits and QA checklist.

Pitfalls and countermeasures

1

Hallucination

Ground claims and require a fact/uncertainty table.

2

Generic content

Add first-party examples and expert review.

3

Keyword stuffing

Optimise for intent and natural coverage, not repetition.

4

Privacy leakage

Use public or synthetic inputs; redact customer data.

5

Copyright risk

Do not imitate living creators or reproduce source text.

6

Automation drift

Version prompts, log changes and monitor outcomes.

Applied Labs — Fundamentals of SEO and Generative AI

Lab 1

- Audit the EcoDesk Demo Website
- LO1 · A6 · K1 · K6

Lab 2

- Design the SIGNAL AI SEO Strategy
- LO1 · A6 · K1 · K6

Safe demo evidence

- Fictional EcoDesk website
- Public guidance and synthetic data clearly labelled

Audit the EcoDesk Demo Website

LAB 1

Establish an evidence-based baseline for a fictional Singapore sustainable-office-products website before any AI-assisted changes.

You'll build

Baseline HTML/CSS/JavaScript site, audit scorecard, issue register and prioritised recommendations

Tools / evidence: Browser developer tools · page source · supplied audit prompt · evidence checklist

Scenario — Evidence and Guardrails

Evidence basis

- EcoDesk is a fictional training business. The baseline intentionally contains realistic SEO strengths and weaknesses so learners can inspect code, content and user experience safely.
- Source:
<https://developers.google.com/search/docs/essentials>

Training assumptions / limits

- All business names, addresses, prices, testimonials and analytics are synthetic.
- No live Search Console or customer data is required.
- The audit must distinguish observed evidence from an AI inference.

Decision Questions

1

Which issues affect crawl/index eligibility, relevance, trust or conversion?

2

Which current strengths should be preserved?

3

Which recommendation has the highest impact-to-effort ratio?

4

What evidence is missing before implementation?

Audit the EcoDesk Demo Website

Test it

Every finding cites observable evidence, separates fact from inference, includes a verification method and maps to a business or user outcome.

Design the SIGNAL AI SEO Strategy

LAB 2

Turn the baseline into a 90-day AI-assisted SEO strategy with outcomes, audiences, content priorities, controls and measurement.

You'll build

Strategy webpage, SIGNAL canvas, 90-day roadmap, prompt policy and agent approval matrix

Tools / evidence: Baseline audit · SIGNAL framework · strategy prompt · governance checklist

Scenario — Evidence and Guardrails

Evidence basis

- Google permits appropriate use of generative AI but warns that scaled, low-value content created mainly to manipulate rankings can violate spam policies.
- Source:
<https://developers.google.com/search/docs/fundamentals/using-gen-ai-content>

Training assumptions / limits

- The 90-day budget is S\$6,000 and eight staff-days.
- Only public and synthetic data may enter general-purpose AI tools.
- Publishing, outreach and technical directives require named human approval.

Decision Questions

1

What outcome should SEO create beyond traffic?

2

Which information advantage can EcoDesk credibly own?

3

Which tasks are suitable for copilot, workflow or agent mode?

4

Which risks require a stop rule or human approval?

Design the SIGNAL AI SEO Strategy

Test it

The strategy links each initiative to an outcome, source of truth, owner, approval gate, metric and rollback or review trigger.

Recap — Fundamentals of SEO and Generative AI

1

Audit the EcoDesk Demo Website

Evaluate a website's current SEO and internet-marketing strategy

2

Design the SIGNAL AI SEO Strategy

Create an organisation-level AI SEO strategy and governed workflow

3

Evidence habit

Separate official guidance, first-party data, training assumptions and AI inference.

4

Management habit

Link every recommendation to an owner, approval gate, metric and review trigger.

LEARNING UNIT 02 · A1 · K2 · K8

AI-Powered Keyword Research and Content Strategy

Intent · entities · clustering · prioritisation · content architecture

Key Concepts — AI-Powered Keyword Research and Content Strategy

1

Demand

Keywords reveal language and questions, not guaranteed traffic.

2

Intent

Classify what the searcher wants to know, compare, do or buy.

3

Topic model

Connect entities, subtopics and follow-up questions into useful coverage.

4

Prioritisation

Balance relevance, business value, difficulty, evidence and effort.

TOPIC 02 · A1 · K2 · K8

A keyword list becomes a strategy only after intent, evidence and priority are added.

AI is strong at expansion and clustering; real search data and business judgement remain the source of truth.

Keyword dimensions

1

Head vs long-tail

Breadth and volume versus specificity and conversion context.

2

Branded vs non-branded

Known demand versus category discovery.

3

Local vs national

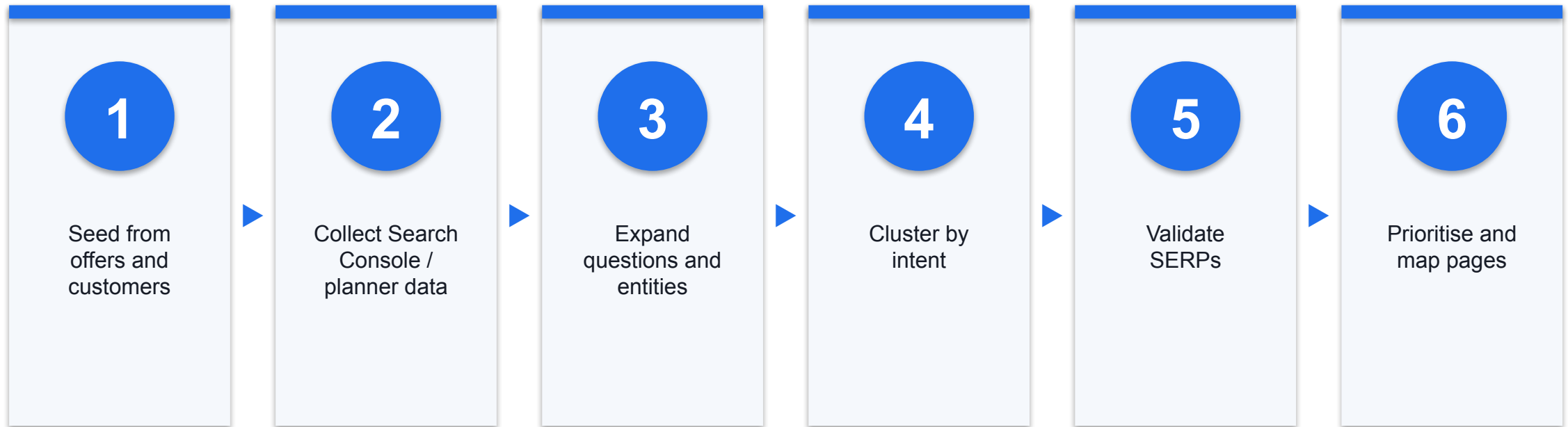
Place, proximity and service-area context.

4

Question vs action

Learning needs versus comparison, booking or purchase.

Grounded AI keyword workflow



Sources of keyword evidence

1

First party

Site search, sales calls, support tickets and Search Console queries.

2

Google surfaces

Autocomplete, People Also Ask, related searches and Trends.

3

Competitive

Ranking pages, content gaps, backlinks and paid-search language.

4

Commercial tools

Planner, Ahrefs, Semrush and alternatives; verify definitions and freshness.

What AI can infer and what it cannot know

Perspective A

- Semantic similarity
- Likely intent
- Question variations
- Entity and modifier patterns

Perspective B

- Current volume without data
- Actual difficulty
- Your conversion rate
- Business priority and legal constraints

Search-intent matrix

1

Know

Definitions, explanations, guides and evidence.

2

Do

Procedures, tools, templates and checklists.

3

Compare

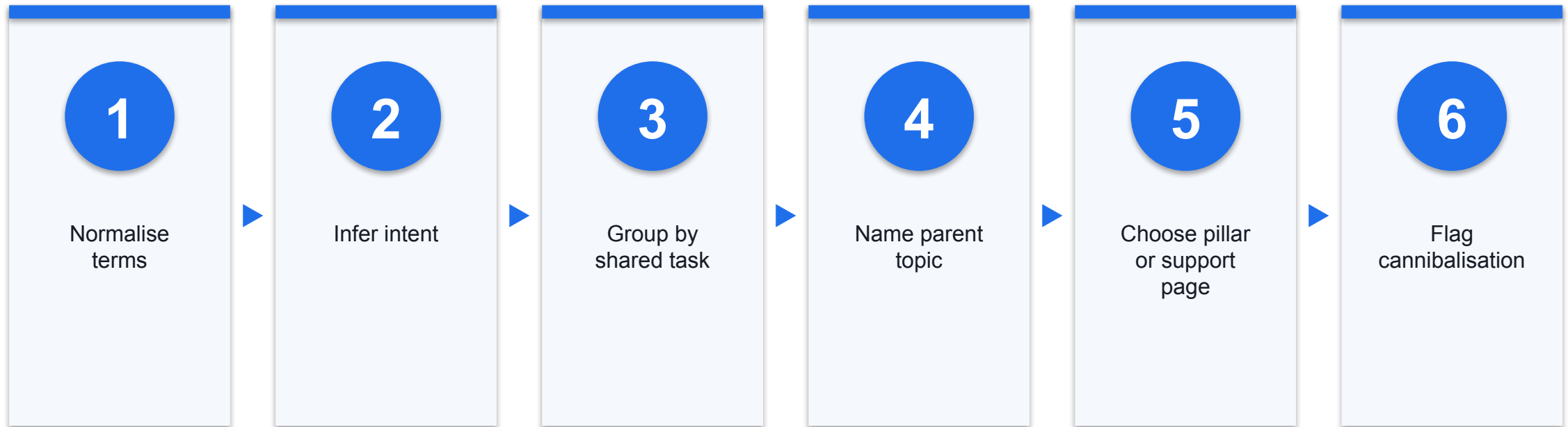
Alternatives, criteria, trade-offs and use cases.

4

Buy / visit

Offer, trust, price, availability, location and action.

Prompted cluster method



Keyword opportunity score

1

Relevance

Does the query fit the offer and audience?

2

Evidence

Can we add credible, differentiated value?

3

Demand

Is there observable interest or strategic importance?

4

Competition

Can the site plausibly earn visibility?

5

Conversion

Does the intent support a valued action?

6

Effort

What content, technical and authority work is needed?

Content architecture

Pillar

- Broad decision guide connecting the topic cluster.

Cluster

- Focused pages resolving sub-intents without duplication.

Conversion

- Service, product or lead page connected by relevant internal links.

Cannibalisation warning signs

1

Same intent

Multiple pages promise the same answer.

2

SERP switching

Different URLs rank for one query over time.

3

Split authority

Links and engagement fragment across near-duplicates.

4

Fix

Consolidate, differentiate purpose, redirect or adjust internal links.

Prompt controls for keyword research

1

Input table

Require keyword, source, metric date and landing page.

2

No invented metrics

Use null/unknown when volume or difficulty is missing.

3

Reason codes

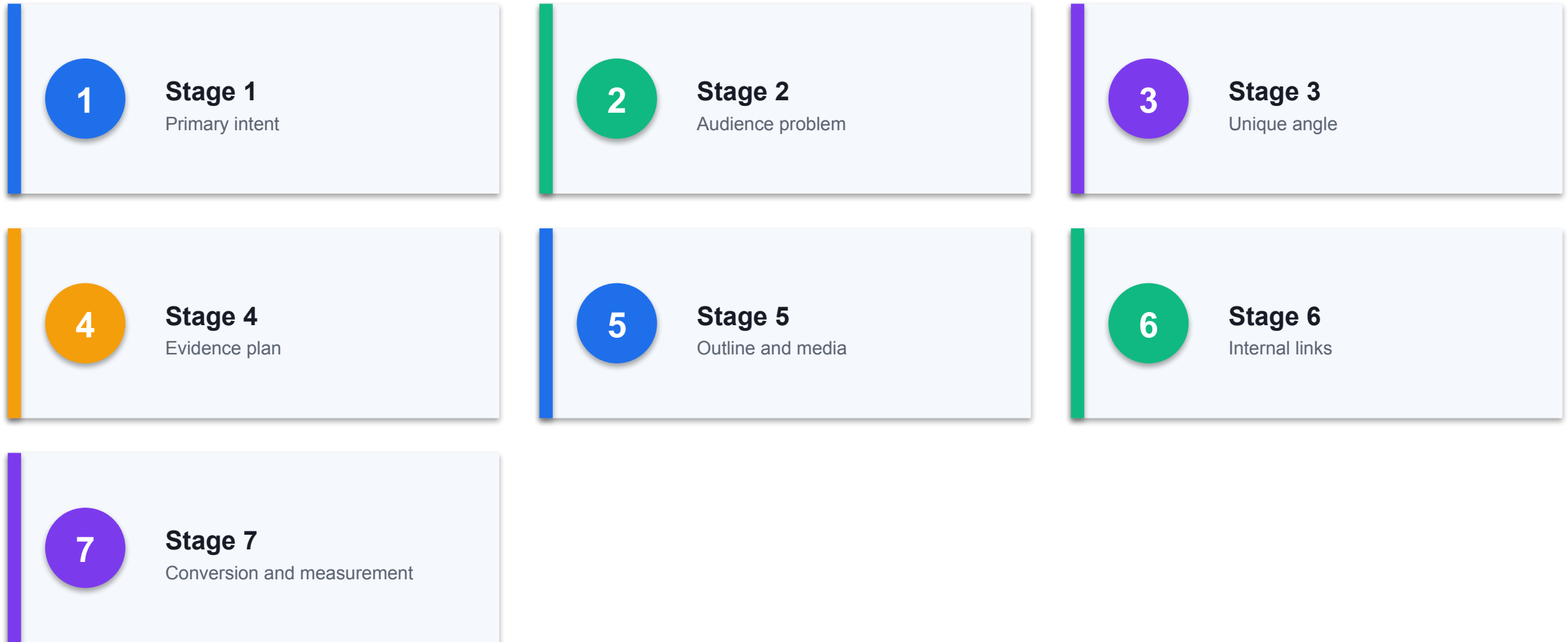
Explain intent and cluster assignment briefly.

4

Review queue

Flag ambiguous, sensitive or low-confidence terms.

From keyword to editorial brief



Applied Labs — AI-Powered Keyword Research and Content Strategy

Lab 3

- Generate and Validate a Keyword Universe
- LO2 · A1 · K2 · K8

Lab 4

- Cluster Keywords and Build Content Briefs
- LO2 · A1 · K2 · K8

Safe demo evidence

- Fictional EcoDesk website
- Public guidance and synthetic data clearly labelled

Generate and Validate a Keyword Universe

LAB 3

Create a grounded keyword universe for EcoDesk from supplied first-party notes and synthetic planner/Search Console exports.

You'll build

Keyword-hub website page, prompt output, cleaned keyword table and validation log

Tools / evidence: Synthetic CSV · keyword prompt · search-intent rubric · browser

Scenario — Evidence and Guardrails

Evidence basis

- AI can expand, normalise and group language, but search volume, competition and current SERP evidence must come from supplied or verified data.
- Source:
<https://support.google.com/google-ads/answer/7337243>

Training assumptions / limits

- Keyword metrics are synthetic and dated 16 August 2026.
- Missing volume and difficulty must remain unknown, never estimated by the model.
- Search intent requires manual SERP validation before publishing.

Decision Questions

1

Which seeds come from real customer language?

2

Which expansions are relevant but unsupported by demand data?

3

Which queries reveal local, comparison or purchase intent?

4

Which terms should be excluded for weak fit or compliance risk?

Generate and Validate a Keyword Universe

Test it

No metric is AI-invented; every retained keyword has a source, intent, cluster candidate, validation status and business-fit rationale.

Cluster Keywords and Build Content Briefs

LAB 4

Transform the validated keyword universe into a non-cannibalising topic cluster and three grounded content briefs.

You'll build

Content-plan webpage, cluster map, page-keyword matrix and three briefs

Tools / evidence: Lab 3 keyword table · clustering prompt · brief template · cannibalisation checklist

Scenario — Evidence and Guardrails

Evidence basis

- Useful information architecture groups content by audience tasks and makes important pages reachable through descriptive internal links.
- Source:
<https://developers.google.com/search/docs/fundamentals/seo-starter-guide>

Training assumptions / limits

- The approved cluster is sustainable office setup for Singapore SMEs.
- EcoDesk has access to an operations specialist for first-party examples.
- Environmental claims require documentary support and cautious wording.

Decision Questions

1

Which terms share the same dominant intent?

2

Which page should be pillar, support or conversion content?

3

Where could two proposed pages cannibalise each other?

4

What unique evidence makes each brief publishable?

Cluster Keywords and Build Content Briefs

Test it

Every keyword maps to one primary page, each brief has a distinct intent and information advantage, and internal links form a logical user journey.

Recap — AI-Powered Keyword Research and Content Strategy

1

Generate and Validate a Keyword Universe

Use AI to expand and validate search terms without inventing metrics

2

Cluster Keywords and Build Content Briefs

Map validated keywords to intent-led pages and editorial briefs

3

Evidence habit

Separate official guidance, first-party data, training assumptions and AI inference.

4

Management habit

Link every recommendation to an owner, approval gate, metric and review trigger.

LEARNING UNIT 03 · A5 · K5 · K7

Creating SEO-Optimised Content with Generative AI

Blogs · articles · landing pages · images · video · social repurposing

Key Concepts — Creating SEO-Optimised Content with Generative AI

1

Brief

A grounded contract for audience, intent, evidence, angle and outcome.

2

Draft

AI proposes structure and language; experts add truth and experience.

3

Media

Images and video answer visual intent and improve comprehension.

4

Distribution

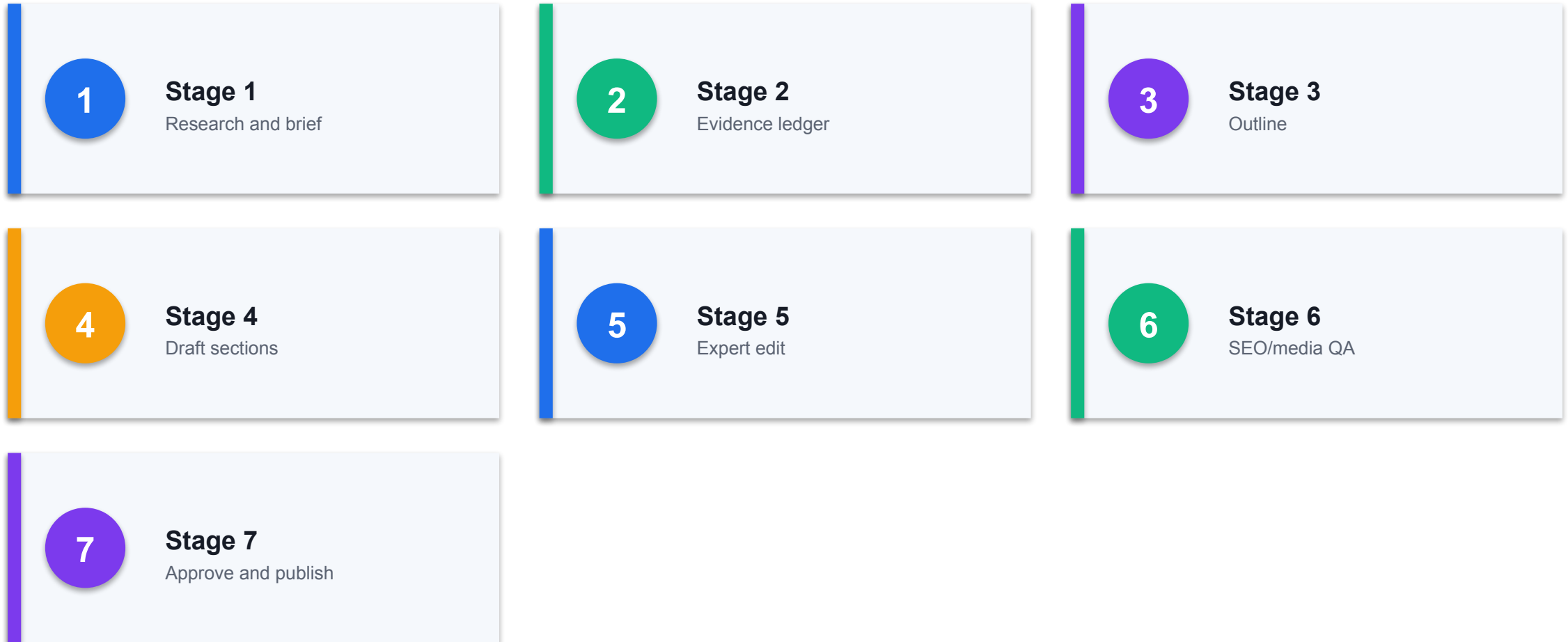
Social content earns attention and relationships; backlinks must be editorially deserved.

TOPIC 03 · A5 · K5 · K7

The publishable unit is not an AI draft; it is a verified, edited asset with a reason to exist.

Original evidence, useful structure and human accountability separate content from commodity output.

Editorial workflow



A strong content brief

1

Audience

Role, context, knowledge and decision.

2

Intent

Primary question and follow-up needs.

3

Angle

Specific promise and information advantage.

4

Evidence

Claims, sources, examples and subject-matter input.

5

Structure

Answer-first headings, comparison, process and FAQ.

6

Action

Next useful step and measurement event.

Answer-first does and don'ts

Perspective A

- Lead with a concise answer
- Use descriptive headings
- Show steps, tables and examples
- Cite facts near the claim

Perspective B

- Do not pad the introduction
- Do not repeat the keyword unnaturally
- Do not hide uncertainty
- Do not promise rankings or AI citations

E-E-A-T as an editorial review lens

1

Experience

Demonstrate first-hand process, examples and outcomes.

2

Expertise

Use qualified authors and technically accurate explanations.

3

Authoritativeness

Earn recognition, references and consistent entity information.

4

Trust

Make sources, authorship, updates, policies and limitations clear.

Fact-safe drafting loop

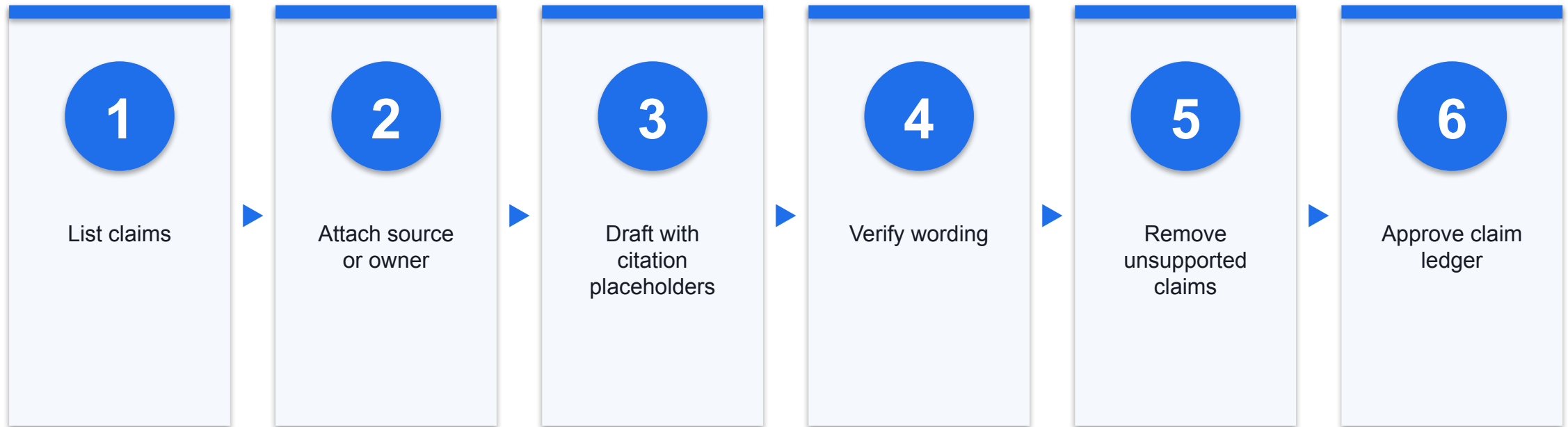


Image-generation prompt controls

1

Purpose

Define query, page section and learning/conversion job.

2

Composition

Specify subject, framing and responsive crop zones.

3

Accuracy

List must-have details and prohibited artefacts.

4

Rights

Avoid logos, copyrighted characters and creator imitation.

5

Delivery

File type, dimensions, compression and descriptive filename.

6

Accessibility

Write alt text for function and context, not keyword stuffing.

Image SEO

1

Relevant original visual

Illustrate the page's actual topic or evidence.

2

Efficient delivery

Responsive sizes, WebP/AVIF where appropriate, dimensions and lazy loading.

3

Discoverable context

Descriptive filename, nearby text and image sitemap where needed.

4

Useful alt text

Concisely communicate purpose to users who cannot see the image.

Video SEO

1

Searchable page

Give each important video a stable, indexable watch page.

2

Transcript

Expose accurate spoken content as useful text.

3

Metadata

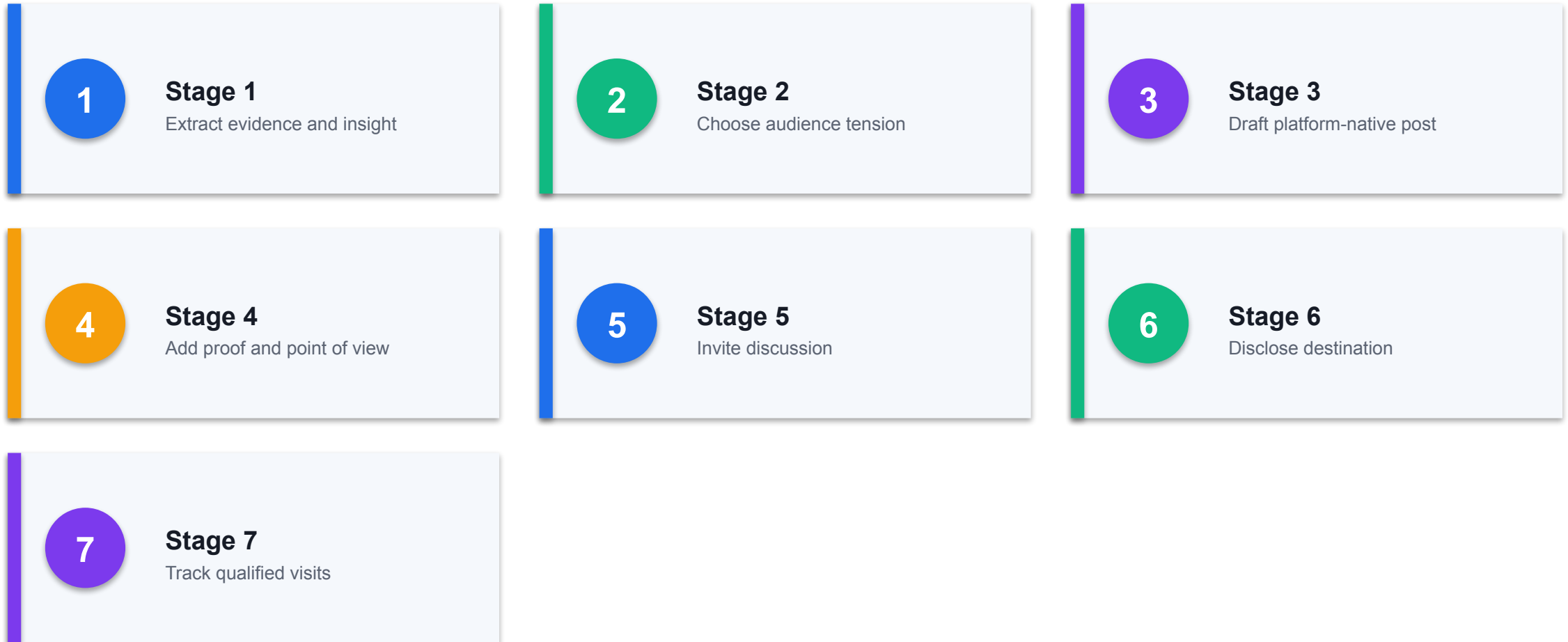
Descriptive title, thumbnail, duration, upload date and VideoObject where valid.

4

Chapters

Help users navigate moments and clarify subtopics.

One article to a social campaign



Earned links and manipulative links

Perspective A

- Original research and tools
- Expert commentary
- Useful resources
- Relevant outreach and digital PR

Perspective B

- Bought or exchanged ranking links
- Mass low-quality guest posts
- Automated comment/forum spam
- Invented endorsements or fake profiles

AI content QA scorecard

1

Intent

Resolves the real task completely.

2

Accuracy

Every material claim is verified.

3

Originality

Adds examples, data or judgement unavailable elsewhere.

4

Clarity

Scannable, specific and free of filler.

5

Brand

Voice, risk and disclosure requirements met.

6

SEO

Metadata, links, media and structured data match the visible page.

Applied Labs — Creating SEO-Optimised Content with Generative AI

Lab 5

- Create a People-First SEO Blog Article
- LO3 · A5 · K7 · K10

Lab 6

- Generate Image, Video and LinkedIn Assets
- LO4 · A3 · A5 · K3 · K5

Safe demo evidence

- Fictional EcoDesk website
- Public guidance and synthetic data clearly labelled

Create a People-First SEO Blog Article

LAB 5

Use a staged prompt workflow to create an answer-first article from a verified brief and claim ledger.

You'll build

Published static blog page, draft comparison, claim ledger, editorial QA and internal links

Tools / evidence: Lab 4 brief · source pack · blog prompts · claim ledger

Scenario — Evidence and Guardrails

Evidence basis

- Helpful content is created primarily to benefit people; AI-assisted content still needs originality, accuracy, transparent authorship and appropriate expertise.
- Source:
<https://developers.google.com/search/docs/fundamentals/creating-helpful-content>

Training assumptions / limits

- All examples are illustrative and must be labelled when not source-backed.
- The article must not claim certification, guaranteed savings or environmental impact without evidence.
- The author is an EcoDesk workplace sustainability adviser with a transparent bio.

Decision Questions

1

What is the reader's main decision?

2

Which claims need sources or qualification?

3

What first-party experience can make the article non-commodity?

4

Which internal links continue the user journey?

Create a People-First SEO Blog Article

Test it

Every material claim is supported or qualified, the page resolves intent, adds original operational value and passes the six-part AI content QA scorecard.

Generate Image, Video and LinkedIn Assets

LAB 6

Prompt text-to-image and video concepts, write accessible metadata, and repurpose the blog into a LinkedIn post designed to earn attention and editorial backlinks.

You'll build

Media-library webpage, image prompt, storyboard, transcript, metadata and LinkedIn outreach post

Tools / evidence: Image/video prompt pack · alt-text rubric · VideoObject checklist · social brief

Scenario — Evidence and Guardrails

Evidence basis

- Images and video should be relevant, accessible and technically efficient. Social distribution can earn discovery and links, but manipulative link schemes remain prohibited.
- Source:
<https://developers.google.com/search/docs/appearance/google-images>

Training assumptions / limits

- Generated visuals may not include logos, copyrighted characters or unsupported certifications.
- The video is a concept storyboard; learners do not need paid generation tools.
- The LinkedIn post must not ask for reciprocal or paid links.

Decision Questions

1

What job does each media asset perform?

2

What visual inaccuracies or rights risks must the prompt prevent?

3

What belongs in alt text versus a caption?

4

Why would another professional voluntarily reference the article?

Generate Image, Video and LinkedIn Assets

Test it

Assets are relevant, accessible, rights-safe and evidence-aligned; the LinkedIn post offers standalone value and seeks earned rather than manipulated links.

Recap — Creating SEO-Optimised Content with Generative AI

1

Create a People-First SEO Blog Article

Generate and edit a grounded long-form article

2

Generate Image, Video and LinkedIn Assets

Create multimodal SEO assets and an ethical off-page distribution plan

3

Evidence habit

Separate official guidance, first-party data, training assumptions and AI inference.

4

Management habit

Link every recommendation to an owner, approval gate, metric and review trigger.

LEARNING UNIT 04 · A2 · K10

AI for On-Page SEO and Content Optimisation

Metadata · architecture · technical SEO · structured data · AI Overviews

Key Concepts — AI for On-Page SEO and Content Optimisation

1

Page signal

Align title, heading, copy, media, links and structured data with one useful purpose.

2

Technical eligibility

Search must be allowed to crawl, render, index and show a snippet.

3

Information architecture

Make priority pages reachable and relationships explicit.

4

AI features

Foundational SEO and unique, people-first content remain the core.

There is no magic AI Overview schema.

A page must first be indexable and eligible for Google Search; clear, unique and reliable content does the rest.

On-page signal stack

1

Snippet

Title and description set expectation; Google may rewrite them.

2

Structure

One clear H1 with descriptive H2/H3 sections.

3

Content

Complete, accurate answer with original evidence.

4

Media

Helpful images/video with accessible context.

5

Links

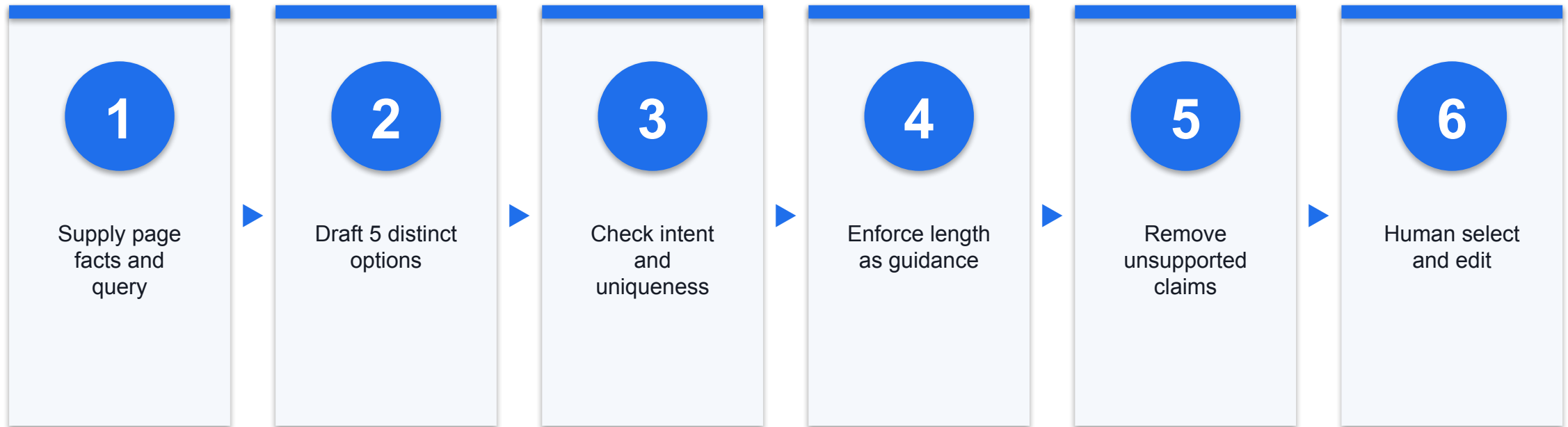
Descriptive internal links and cited external evidence.

6

Markup

Structured data that matches visible content.

Metadata prompt workflow



Title and description quality

1

Specific

Name the page's value and distinguishing context.

2

Accurate

Do not promise content or offers absent from the page.

3

Unique

Avoid sitewide boilerplate and near duplicates.

4

Readable

Natural language first; no keyword lists or excessive punctuation.

URL, canonical and index controls

1

URL

Stable, concise, descriptive and consistent.

2

Canonical

Indicate preferred duplicate while keeping internal signals aligned.

3

Noindex

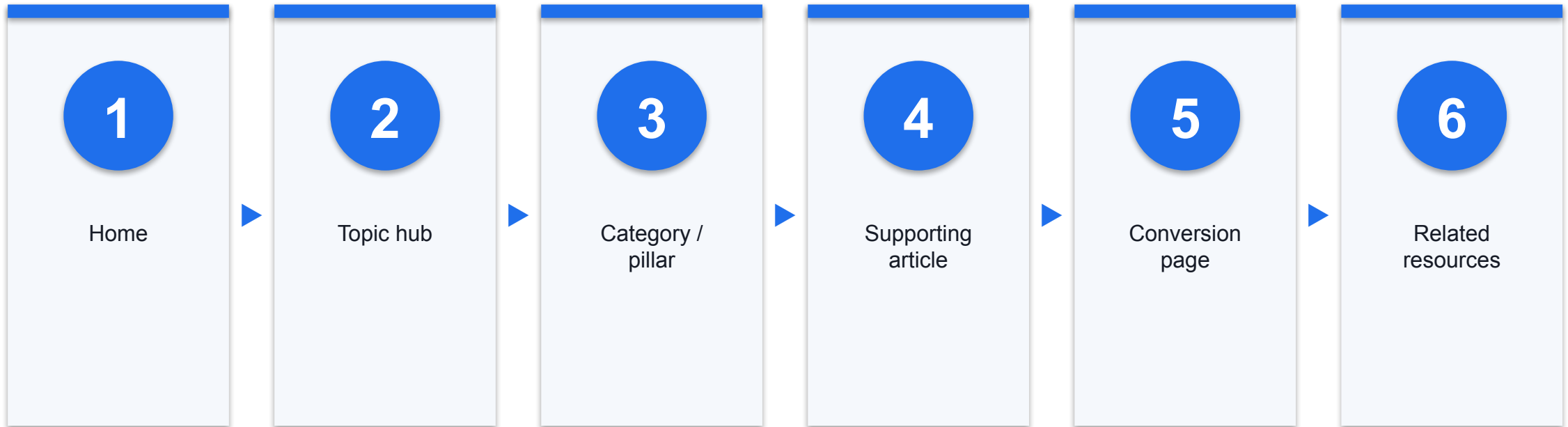
Remove a page from search; do not rely on robots.txt for this.

4

Robots

Control crawling, not guaranteed indexing removal.

Site architecture



Internal-link decisions

1

Relevance

Link where the destination helps the reader continue.

2

Anchor

Describe the destination naturally.

3

Priority

Make valuable pages reachable in few logical steps.

4

Maintenance

Repair broken, redirected and orphaned links.

Structured data rules

1

Eligible type

Use a Google-supported type only when the page qualifies.

2

Visible match

Markup must reflect content users can see.

3

Required properties

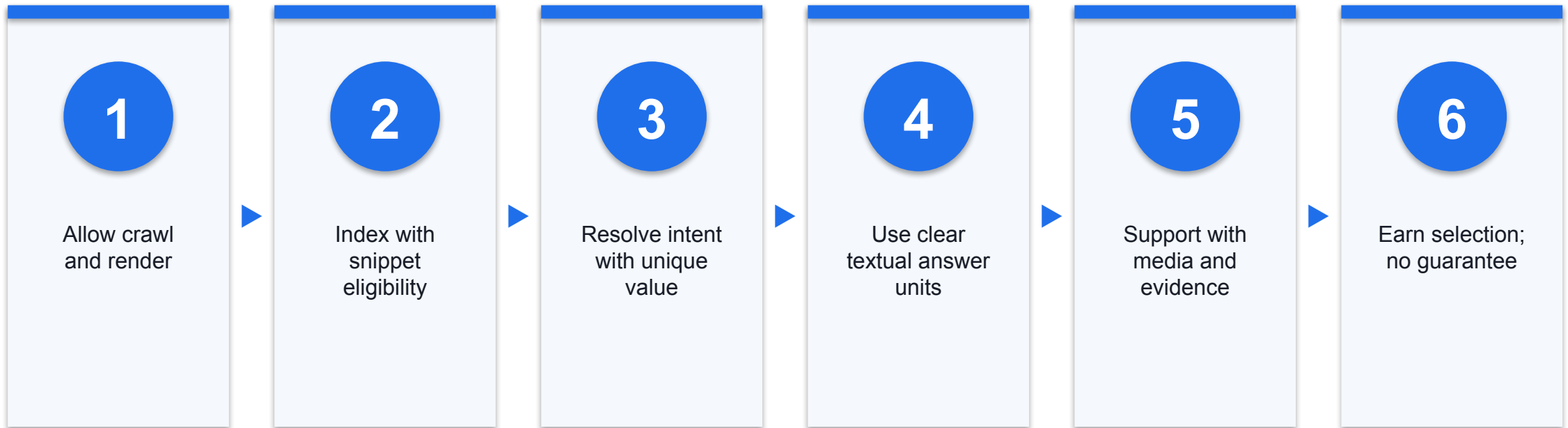
Include valid, current values and stable URLs.

4

Validation

Test syntax and monitor enhancements; rich results are not guaranteed.

AI Overview / AI Mode eligibility



Query fan-out implications

1

Subtopics

Cover the natural supporting questions without padding.

2

Entities

Use consistent names and relationships.

3

Comparisons

Provide criteria, trade-offs and decision context.

4

Freshness

Update material facts and show meaningful review dates.

Useful GEO/AEO practice and unsupported hacks

Perspective A

- Foundational technical SEO
- People-first, non-commodity content
- Clear structure and first-party evidence
- Search Console measurement

Perspective B

- Special AI-only markup
- Guaranteed citation formulas
- Inauthentic mentions
- Publishing an llms.txt file as a Google ranking tactic

Core Web and mobile hygiene

1

LCP

Deliver the main content quickly with efficient media.

2

INP

Keep interaction responsive by controlling script work.

3

CLS

Reserve space for images, embeds and dynamic UI.

4

Mobile

Readable, usable and equivalent primary content across devices.

Applied Labs — AI for On-Page SEO and Content Optimisation

Lab 7

- Generate and Implement On-Page SEO Meta
- LO3 · A2 · K10

Lab 8

- Optimise an Answer Page for Google AI Overviews
- LO3 · A2 · K4 · K10

Safe demo evidence

- Fictional EcoDesk website
- Public guidance and synthetic data clearly labelled

Generate and Implement On-Page SEO Meta

LAB 7

Generate metadata options from visible page facts, then implement HTML, internal links, image attributes and valid structured data.

You'll build

Optimised products page with meta, canonical, headings, internal links, images and JSON-LD

Tools / evidence: Product facts · metadata prompt · HTML validator · rich-results checklist

Scenario — Evidence and Guardrails

Evidence basis

- Titles and snippets should accurately represent visible content. Structured data must match the page and does not guarantee a rich result.
- Source:
<https://developers.google.com/search/docs/appearance/title-link>

Training assumptions / limits

- Product names and descriptions are fictional.
- No reviews, aggregate rating or certification markup may be added.
- The canonical URL is a training placeholder and must be changed before real deployment.

Decision Questions

1

Which page facts can safely appear in metadata?

2

Which generated option best matches intent and brand?

3

Which structured-data type is eligible and visible?

4

Which internal links reduce user effort?

Generate and Implement On-Page SEO Meta

Test it

Metadata is unique and accurate, visible content and JSON-LD agree, links work, images are accessible and no unsupported rating, price or guarantee is present.

Optimise an Answer Page for Google AI Overviews

LAB 8

Create a clear, source-grounded page that can be indexed and understood by users and generative search features without relying on unsupported GEO hacks.

You'll build

AI Overview-ready guide, answer units, comparison table, FAQ, sources and technical eligibility checklist

Tools / evidence: Official Google guidance · query fan-out prompt · answer-unit rubric · source ledger

Scenario — Evidence and Guardrails

Evidence basis

- Google says existing SEO best practices remain relevant for AI features, pages need no special AI markup, and inclusion is never guaranteed.
- Source:
<https://developers.google.com/search/docs/fundamentals/ai-optimization-guide>

Training assumptions / limits

- The page targets Singapore SME office managers.
- The guide uses public general information and synthetic EcoDesk examples.
- FAQ structured data is omitted unless current eligibility and visible-content requirements are met.

Decision Questions

1

What direct answer should appear first?

2

Which follow-up subqueries might query fan-out retrieve?

3

What original evidence or experience improves the page?

4

Which technical controls could accidentally remove eligibility?

Optimise an Answer Page for Google AI Overviews

Test it

The page is indexable, resolves the main and follow-up intents with unique value, cites evidence and makes no guarantee of AI Overview inclusion.

Recap — AI for On-Page SEO and Content Optimisation

1

Generate and Implement On-Page SEO Meta

Use prompts to create and implement accurate on-page elements

2

Optimise an Answer Page for Google AI Overviews

Apply foundational SEO and answer-first content for generative search

3

Evidence habit

Separate official guidance, first-party data, training assumptions and AI inference.

4

Management habit

Link every recommendation to an owner, approval gate, metric and review trigger.

LEARNING UNIT 05 · A3 · A4 · K3 · K4 · K9

SEO Performance, Ethics and Continuous Improvement with AI

Search Console · GA4 · AI visibility · experiments · governance

Key Concepts — SEO Performance, Ethics and Continuous Improvement with AI

1

Visibility

Search impressions and cited presence show discoverability, not business value alone.

2

Engagement

Qualified visits and useful actions indicate whether intent was met.

3

Conversion

Connect organic journeys to leads, sales or other valued outcomes.

4

Learning

Use baselines, hypotheses and controlled changes before attributing impact.

TOPIC 05 · A3 · A4 · K3 · K4 · K9

AI can explain a dashboard; it cannot repair weak measurement.

Clean definitions, complete inputs and cautious attribution come before generated insights.

Measurement stack

1

Search Console

Queries, pages, clicks, impressions, CTR, position and indexing.

2

GA4

Sessions, users, engagement, events, key events and landing-page outcomes.

3

Technical crawl

Status, metadata, canonicals, links, directives and performance.

4

Business system

Lead quality, revenue, margin, retention and offline outcome.

Search Console and GA4 answer different questions

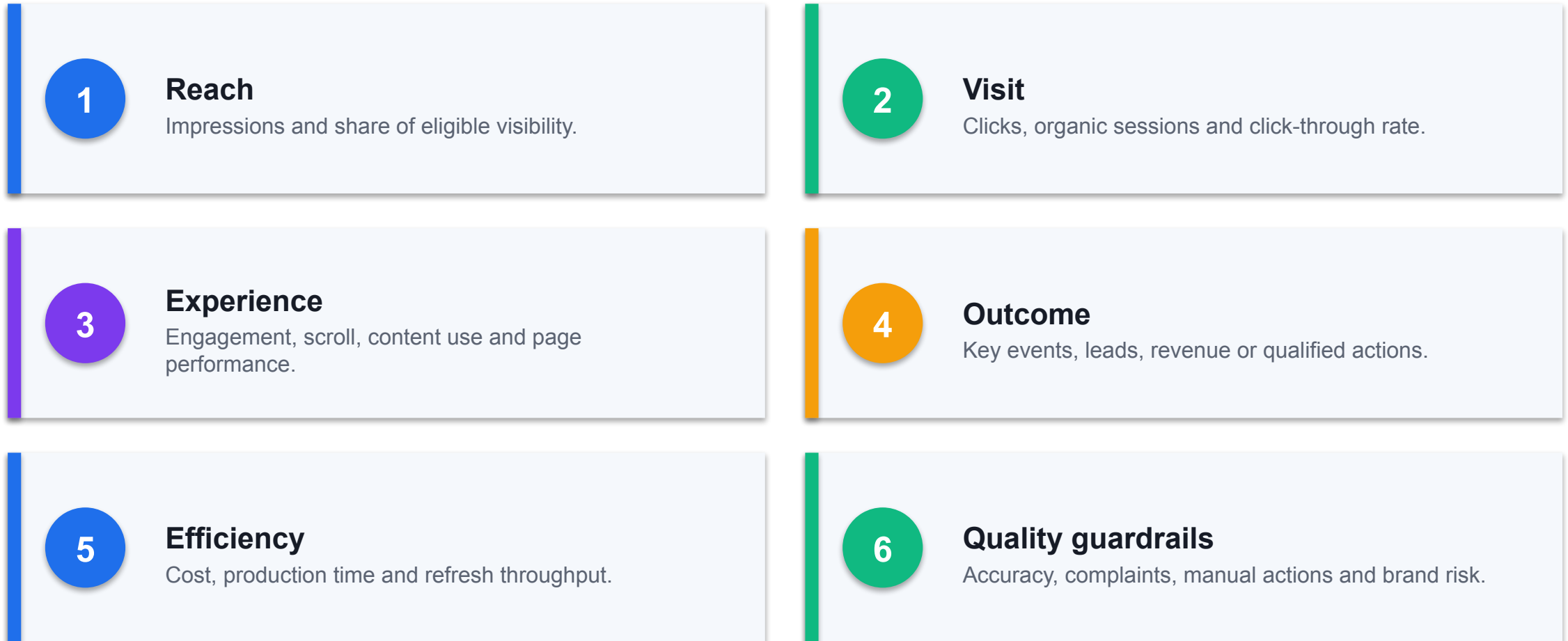
Perspective A

- Did Google show and send clicks?
- Which query/page combinations changed?
- Is the URL indexed and eligible?
- Which search features are present?

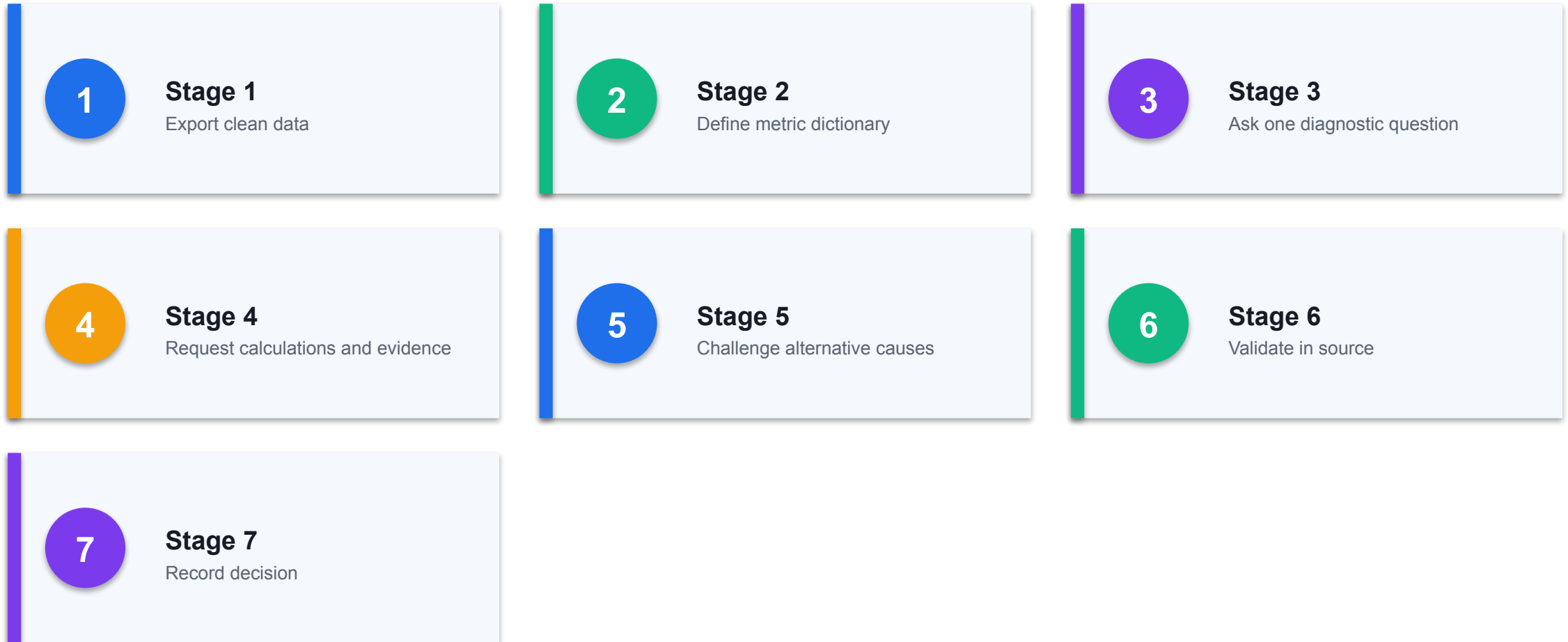
Perspective B

- What did visitors do after arrival?
- Which landing pages drove key events?
- How do channels and campaigns compare?
- What is the user/session journey?

SEO KPI tree



AI-assisted analysis



Avoid attribution traps

1

Seasonality

Compare relevant periods and known demand cycles.

2

SERP change

AI features, ads and layout can change clicks without ranking loss.

3

Tracking change

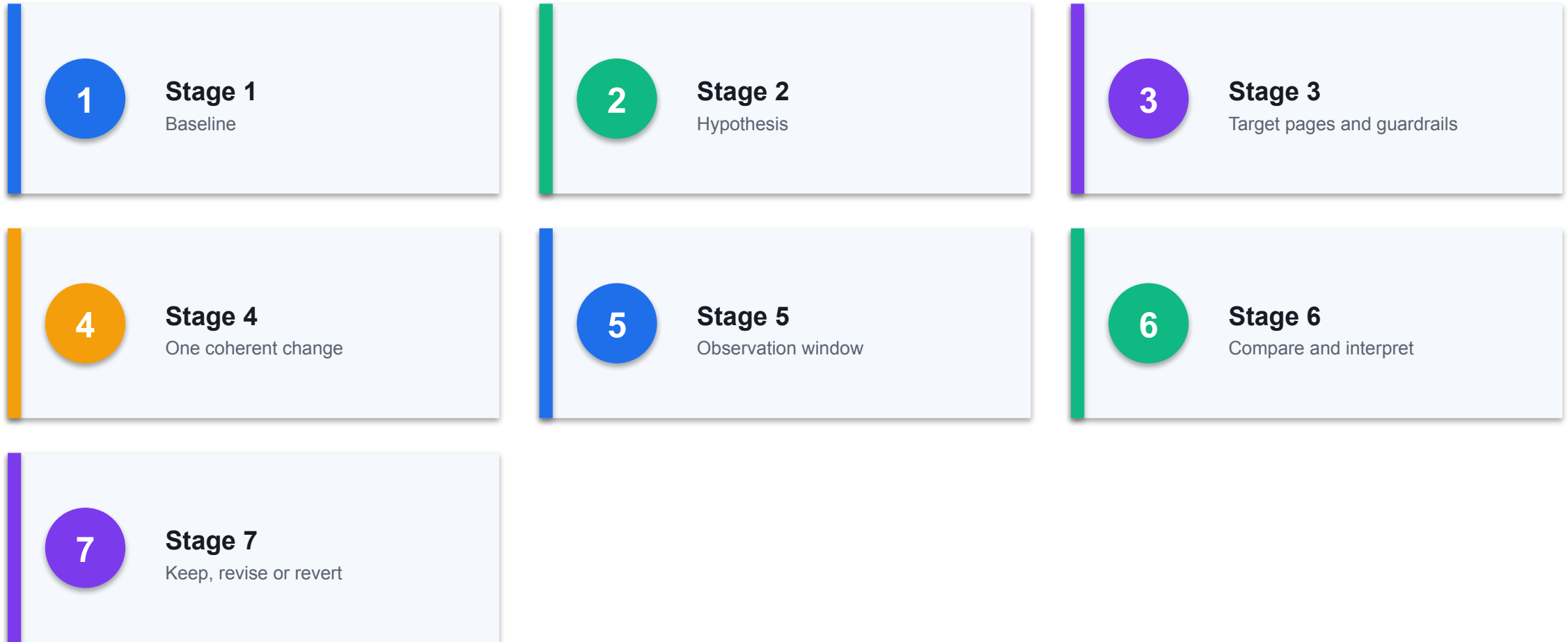
Consent, tagging and definitions can create artificial shifts.

4

Content mix

Averages hide landing-page and intent differences.

Experiment discipline



AI visibility measurement

1

Official

Use Search Console reports where available and overall performance data.

2

Manual

Document query, locale, device, date and cited sources.

3

Third-party

Treat estimates as directional and check methodology.

4

Business

Measure AI-referred engagement and conversion where identifiable.

Agentic monitoring controls

1

Read-only first

Let agents detect and recommend before they can edit.

2

Threshold

Require material change and minimum evidence before alerts.

3

Approval

Human approves content, technical and outreach actions.

4

Audit

Log input, model, prompt, output, reviewer, decision and rollback.

Refresh decisions

Maintain

- Content remains accurate, useful and aligned.

Improve

- Update evidence, intent coverage, media, links or conversion path.

Consolidate / retire

- Merge overlap, redirect obsolete pages or noindex when appropriate.

Responsible AI checklist

1

Privacy

No personal, confidential or licensed data without authority.

2

Accuracy

Verify claims, calculations and schema against sources.

3

Fairness

Check stereotypes, exclusions and inappropriate targeting.

4

Transparency

Document AI use where policy or audience trust requires it.

5

Ownership

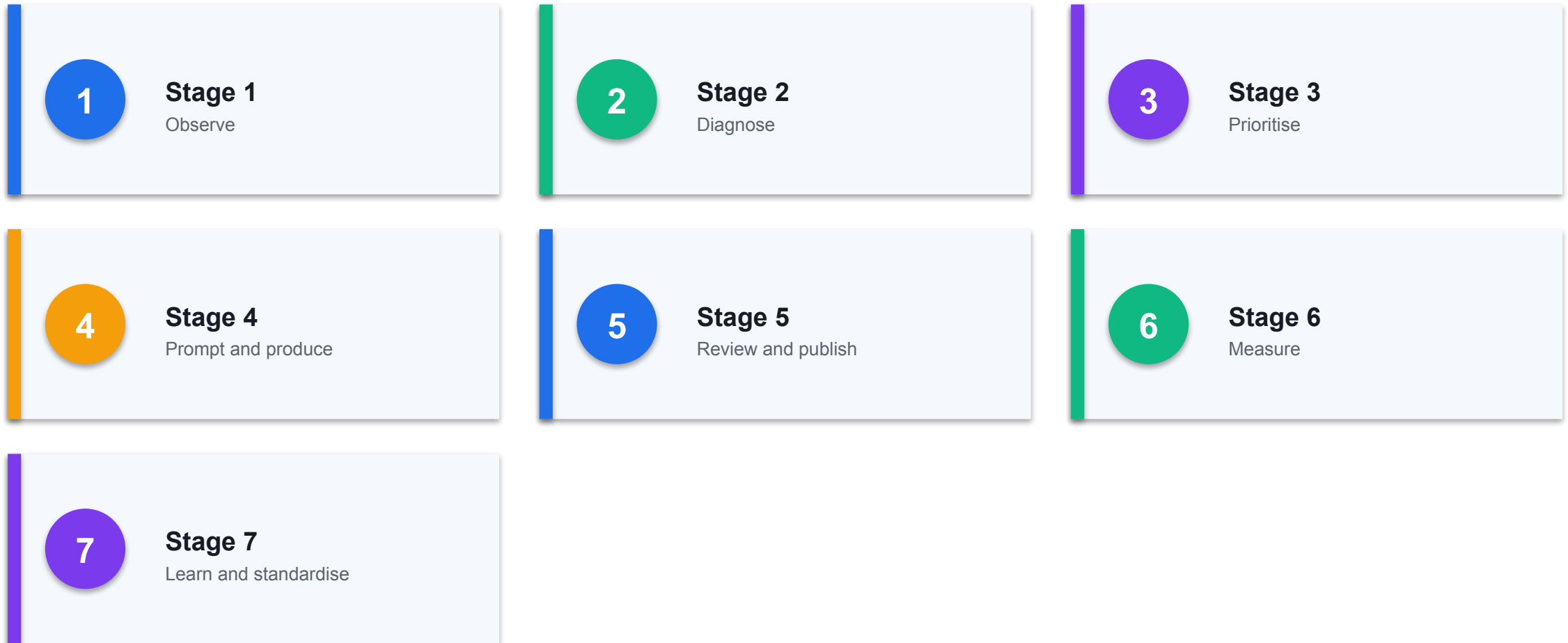
Named reviewer and publisher remain accountable.

6

Security

Least privilege, secret handling and safe external actions.

Continuous improvement loop



Applied Labs — SEO Performance, Ethics and Continuous Improvement with AI

Lab 9

- Analyse Search Console and GA4 Data with AI
- LO5 · A3 · A4 · K3 · K4 · K9

Lab 10

- Build a Governed Agentic SEO Workflow
- LO1–LO5 · A1–A6 · K1–K10

Safe demo evidence

- Fictional EcoDesk website
- Public guidance and synthetic data clearly labelled

Analyse Search Console and GA4 Data with AI

LAB 9

Use synthetic Search Console and GA4 exports to diagnose performance without confusing impressions, clicks, sessions, users or conversions.

You'll build

Performance-review webpage, KPI tree, diagnostic memo, hypotheses and experiment plan

Tools / evidence: Synthetic GSC/GA4 CSV · metric dictionary · analysis prompt · attribution checklist

Scenario — Evidence and Guardrails

Evidence basis

- Search Console describes Google search visibility and clicks; GA4 describes user/session behaviour after arrival. Their scopes and counts need not match.
- Source:
<https://support.google.com/analytics/answer/14731736>

Training assumptions / limits

- All metrics are synthetic and contain a planned tracking change in week 5.
- No causal claim may be made from correlation alone.
- AI calculations must be checked against spreadsheet formulas.

Decision Questions

1

Which metrics changed materially and at what page/query level?

2

Could definition or tracking changes explain the pattern?

3

Which hypotheses fit all available evidence?

4

What test would distinguish competing explanations?

Analyse Search Console and GA4 Data with AI

Test it

Every conclusion traces to supplied data, metric scopes are correct, alternative explanations are considered and the recommended test has a baseline and decision rule.

Build a Governed Agentic SEO Workflow

LAB 10

Integrate the course into a capstone workflow that monitors, recommends and prepares changes while humans retain control of publishing and outreach.

You'll build

Governance webpage, agent workflow, tool permissions, prompt contract, audit log and 30-day monitoring plan

Tools / evidence: All prior lab outputs · workflow canvas · risk matrix · audit-log template

Scenario — Evidence and Guardrails

Evidence basis

- Agentic SEO can coordinate multi-step research, audit and monitoring, but high-risk actions need scoped permissions, approvals, logs and rollback.
- Source:
<https://developers.google.com/search/docs/essentials/spam-policies>

Training assumptions / limits

- The agent has read-only access to analytics and repository content.
- It cannot publish, send outreach, edit robots/canonicals or expose secrets.
- Every recommendation must cite a page, metric row or policy source.

Decision Questions

1

What is the bounded goal and stop condition?

2

Which tools and data does the agent actually need?

3

Which actions need preview, approval or prohibition?

4

How will quality, drift, value and incidents be measured?

Build a Governed Agentic SEO Workflow

Test it

The workflow is evidence-bound, read-only by default, blocks high-risk autonomous actions, records decisions and demonstrates a successful stop/rollback tabletop.

Recap — SEO Performance, Ethics and Continuous Improvement with AI

1

Analyse Search Console and GA4 Data with AI

Evaluate SEO performance across channels using defined metrics

2

Build a Governed Agentic SEO Workflow

Design a monitored AI SEO workflow with approvals and rollback

3

Evidence habit

Separate official guidance, first-party data, training assumptions and AI inference.

4

Management habit

Link every recommendation to an owner, approval gate, metric and review trigger.

WRAP-UP

Visibility Earned Through Useful, Verifiable Work

What You Achieved

1

LO1

Evaluate SEO and internet marketing strategies of a website.

2

LO2

Manage keyword research in alignment with SEO objectives.

3

LO3

Provide on-page SEO strategies and recommendations.

4

LO4

Manage off-page SEO strategies and performance across multiple channels.

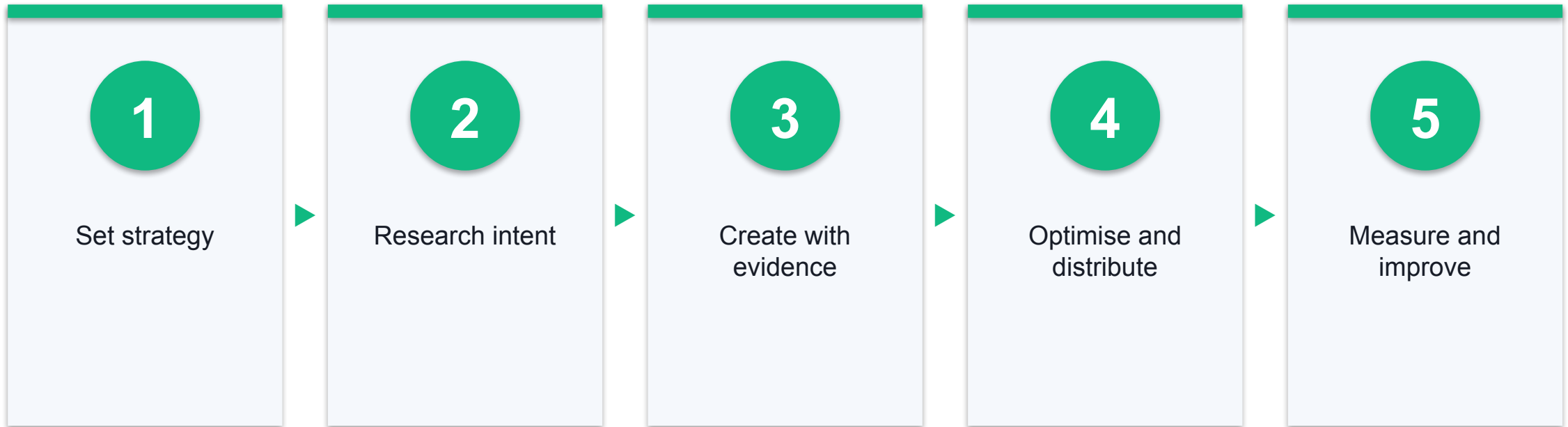
5

LO5

Monitor Google Analytics and Search Console reports to evaluate SEO performance.

TAKEAWAY

The complete AI SEO discipline



Evidence Sources

1

Google: AI features and your website

<https://developers.google.com/search/docs/appearance/ai-features>

2

Google: Optimizing for generative AI features

<https://developers.google.com/search/docs/fundamentals/ai-optimization-guide>

3

Google: Guidance on generative AI content

<https://developers.google.com/search/docs/fundamentals/using-gen-ai-content>

4

Google: Spam policies

<https://developers.google.com/search/docs/essentials/spam-policies>

5

Google Search Essentials

<https://developers.google.com/search/docs/essentials>

6

Google structured data guidelines

<https://developers.google.com/search/docs/appearance/structured-data/sd-policies>

Evidence Sources — Continued

1

Google Search Console performance

<https://support.google.com/webmasters/answer/7576553>

2

Google Analytics acquisition

<https://support.google.com/analytics/answer/14731736>

3

Conductor: AI Overview strategies

<https://www.conductor.com/academy/optimization-strategies-google-ai-overviews/>

4

AllAble: Agentic SEO

<https://www.allable.ai/blog/what-is-agentic-seo/>

5

Authority: Agentic SEO

<https://www.authority.com/blog/agentic-seo>

6

Opinly: Generative AI for SEO

<https://opinly.ai/blog/using-generative-ai-for-seo>

Evidence Sources — Continued

1

Gradient Group: SEO prompt examples
<https://gradientgroup.com/28-expert-level-prompt-examples-for-seo/>

2

Techpresso: SEO prompts
<https://academy.techpresso.co/prompts/chatgpt-prompts-seo>

3

User-supplied source register
<https://www.tertiarycourses.com.sg/generative-ai-for-seo.html>

Share Your Feedback

1

Open

Use the feedback link shared by the trainer.

2

Rate

Tell us what worked well for you.

3

Comment

Suggest what could be improved.

4

Submit

Send it before you leave the session.

GENERATIVE AI FOR SEO

Thank You

Use AI to amplify useful evidence, not to manufacture relevance.